

Rahul Yadav

+1(716)238-0835 | rahulzyadav87@gmail.com | [linkedin.com/in/rahulyad011](https://www.linkedin.com/in/rahulyad011) | [Github](#) | [Website](#) | [Scholar](#) | [ORCID](#)

EXPERIENCE

Oracle

Aug 2024 – Present

Senior Data Scientist (AI/ML Engineering) | Construction Intelligence Team

Malvern, PA

ConstructIQ: Multi-Agent AI Platform for Oracle Construction Cloud

- Architected multi-agent system unifying 5 data modalities (P6, ERP, contract/BIM, CCTV, audio) for Oracle Construction Cloud
- Shipped NEXUS orchestrator on OpenAI Agents SDK coordinating 7 specialist sub-agents with Apache AGE knowledge graph
- Built RAG agent on Oracle 23ai DB across contracts, RFIs, and BIM with hybrid BM25 + vector search at <200ms P95 latency
- Streamed vision events via Kafka consumers into NEXUS knowledge graph, correlated against P6 activity baseline
- Established LangSmith LLM-as-judge eval harness gating CI/CD across all 8 agents on quality, latency, and cost
- Early results with research-partner pilot: ~15% fewer overruns and reporting cut from 40 to 8 hrs/month

Hilabs

Feb 2024 – May 2024

Senior Data Scientist | Applied Data Science Team

Washington, D.C.

LLM-Powered Contract Intelligence for Healthcare Claim Adjudication

- Delivered document understanding system extracting parties, terms, and pricing rules across 5+ insurance contract types
- Achieved 87% F1 on entity extraction by building custom LayoutLM pipeline for joint text/table/image processing
- Deployed self-hosted fine-tuned Mistral on private VPC and retrieval on OpenSearch, meeting HIPAA and PHI requirements
- Delivered V1 to 3 enterprise insurance clients, validating pricing accuracy and fit with their claim processing applications

Apexanalytix

May 2023 – Feb 2024

Data Scientist (Intern to FTE) | Analytics Team

Remote, New York

Enterprise RAG Assistant for Internal & Client Knowledge Access

- Achieved 89% human-eval approval by building production RAG chatbot on LangChain, ChromaDB, and OpenAI generation
- Lifted retrieval relevance by 37% over flat-chunk baseline on internal eval set via parent-child chunking with MMR scoring
- Built feedback collection pipeline driving iterative tuning of prompts, retrieval, and reranking based on real user query patterns
- Drove adoption across 4 internal and 5 external client applications, replacing manual document search across the org

Oracle

Sep 2020 – Aug 2022

Senior ML Engineer | Oracle Financial ML Team

Bengaluru, India

Preemptive Anomaly Detection for Corporate Billing (ORMB)

- Hit 92% precision on 1.2M bills and 5M segments in 20 min via in-memory multivariate detection on Oracle in-database ML
- Built semi-supervised classifier with local + global model explainability for analyst-facing audit and override workflows
- Shipped into Oracle ORMB, deployed across banking, insurance, and telco customers running the platform globally
- Inventor on US Patent 12,333,585 [US12333585B2] for anomaly detection methodology applied to enterprise billing

Wipro Limited

Jun 2018 – Sep 2020

AI/ML Engineer | AI Research Team

Bengaluru, India

NLP Ticketing Assistant for Employee Support Operations

- Cut human-agent intervention by 74% and wait time from 18 to 4 min via BERT-powered semantic search over ticket corpus
- Hit 79% accuracy across 10+ ticket categories via intent classifier and resolution retrieval, powering auto-routing in production

PATENTS & PUBLICATIONS

5 Granted · 4 Pending · 1 Paper

Inventor on 9 USPTO filings spanning anomaly detection, NLP topic categorization, document AI, and multimodal emotion recognition.

Granted

- US 11,386,712 - Multimodal Analysis Based Emotion Recognition [link]
- US 12,002,091 - Technology System for Assisting Financial Institutions in Debt Collection [link]
- US 12,333,585 - Anomaly Detection for Bill Generation (productized in Oracle ORMB) [link]
- US 12,443,792 - Reference Driven NLP-Based Topic Categorization [link]
- US 12,340,613 - Framework for Identifying Documents [link]

Pending

- US App. 17/851,267 - ML Pipeline with Data Quality Framework [link]
- US App. 18/056,456 - Predicting Record Topic Using Transitive Relations [link]
- US App. 17/989,202 - Hierarchical Visualization of Clustered Datasets [link]
- US App. 19/320,494 - Reference Driven NLP-Based Topic Categorization(continuation of US 12,443,792) [link]

Publication

- Virtual Conversation with Real-Time Body Movement / Gesture Prediction - ICMLIP 2019 [link]

EDUCATION

University at Buffalo, SUNY

Aug. 2022 – Dec. 2023

Master of Science, Computer Science and Engineering – CGPA 3.82 / 4.0

Buffalo, NY, USA

- Coursework: NLP, Machine Learning, Pattern Recognition, Deep Learning, Reinforcement Learning, Computing Systems

Indian Institute of Technology (BHU), Varanasi

Jul. 2014 – May 2018

Bachelor of Technology, Electronics Engineering

Varanasi, India

PROJECTS

A2A + MCP Multi-Agent Workflow [\[code\]](#)

2025 – 2026

- Built a production-grade A2A + MCP multi-agent template orchestrating RAG, image-captioning, and ML-classification agents
- Designed a base-class external-agent framework that wraps any external API or model as a plug-and-play A2A agent
- Open-sourced reference coordinating **4** specialist agents end-to-end via A2A handoffs and MCP tool-sharing

Dynamic Agent Spawning Framework [\[code\]](#)

2026

- Built a runtime agent-spawning framework implementing **6** architectures (template-based, LLM-generated, compositional, meta, hierarchical, evolutionary), each swappable via config through a shared orchestrator, tool registry, and base interfaces
- Added a performance tracker capturing execution time, token usage, cost, agents spawned, and success rate per run
- Benchmarked all **6** architectures, finding template-based **~4x** cheaper than LLM-generated at comparable success rate

Doc Generation & Quality Review Agent [\[code\]](#)

2025 – 2026

- Built an LLM tool generating **5** documentation types (README, codeflow, architecture, API, setup) from any codebase
- Designed a quality scorer rating docs across completeness, accuracy, and usability, with an approval workflow gating release
- Added git-diff incremental mode and multi-provider LLM support (OpenAI, Gemini, OCI GenAI) behind one interface

Network-Based Intrusion Detection System (NIDS) [\[code\]](#) [\[results\]](#)

Research Assistant, Dr. Hongxin Hu | Spring 2023

- Engineered intrusion-detection system using deep neural detectors for real-time identification of network threats
- Benchmarked 4 deep neural detector architectures across 12 attack datasets, mapping detection limits per attack type

From RAG POC to Production: What Actually Changes [\[blog\]](#)

Technical Blog · 2026

- Lessons from shipping production RAG: hybrid retrieval beats vector-only, hallucinations are a product problem not a model problem, observability must be built before launch, and trust is won in the first thousand answers
- Covers tiered model selection (50% cost cut at equivalent quality), semantic caching with 30–50% hit rates, citation-first UX patterns, and per-stage observability for retrieval, faithfulness, and latency

SKILLS

Agentic & LLM Systems : OpenAI Agents SDK, LangGraph, LangChain, LlamaIndex, MCP, A2A, RAG, GraphRAG, Tool-use Agents, Function Calling, Prompt Engineering, LLM-as-Judge, LoRA / QLoRA Fine-tuning, LangSmith, Evaluation Harnesses

ML & NLP : PyTorch, TensorFlow, Hugging Face, scikit-learn, Whisper, LayoutLM, Sentence Transformers, Embeddings & Re-rankers, Quantization & Distillation, MLflow, Optuna

Vision : YOLOv11, OpenCV, CLIP, GPT-5.4, RTSP Processing, Object/PPE Detection, Scene Understanding, Segmentation

Data, Vector & Streaming : Kafka, PySpark, Snowflake, SQL, MongoDB, Oracle 23ai Vector DB, Weaviate, FAISS, Apache AGE

Cloud, MLOps & Backend : OCI, AWS, Azure, GCP, Kubernetes, Docker, Terraform, FastAPI, CI/CD, Prometheus, WebSockets

Programming Languages : Python, SQL, JavaScript, Java